Department of Mathematics and Computer Science
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Jonas Henkel

Curriculum Vitae

Studies

From 10/2022 **PhD Student in Mathematics**, *Philipps-University*, Marburg.
Topic: Spectral Geometry on Homogeneous Spaces. Advisor: Prof. Dr. habil. Ilka Agricola

06/2021–09/2022 **Master of Science in Mathematics**, *Philipps-University*, Marburg, Final grade 1.1 (Excellent).
Thesis: The Laplacian on Riemannian Manifolds with Symmetries

10/2015–05/2021 **First State Examination for the Teaching Profession at High Schools in Mathematics and Physics** (Equivalent to M.Ed.), *Philipps-University*, Marburg, Final grade 1.2 (Excellent).
Included comprehensive training in mathematics didactics

Publications

2026 **Explicit Laplace Spectra of Homogeneous Principal Bundles**, *Ilka Agricola, Jonas Henkel*, arXiv:2605.11177v1 [math.DG].
Submitted for review to The Journal of Geometric Analysis

2026 **Hodge Laplacian on 1-forms of homogeneous 3-spheres**, *Jonas Henkel, Emilio Lauret*, arXiv:2605.05406 [math.DG].
Submitted for review to Communications in Analysis and Geometry

2026 **A Machine Learning Approach to the Nirenberg Problem**, *Cortés, Esteban-Casadevall, Feng, Henkel, Hirst, Schettini-Gherardini, Stapleton*, arXiv:2602.12368 [cs.LG].
Submitted for review to Journal of the London Mathematical Society

2025 **The Laplace-Beltrami Spectrum on Naturally Reductive Homogeneous Spaces**, *Ilka Agricola, Jonas Henkel*, arXiv:2503.21416 [math.DG].
Submitted for review to Mathematische Zeitschrift

2025 **The Mathematician's Assistant: Integrating AI into Research Practice**, *Jonas Henkel*, Mathematische Semesterberichte, vol. 72, 117-144.
doi.org/10.1007/s00591-025-00400-0

Computational Mathematics & Programming

Symbolic Computation: Extensive use of Python and SageMath to translate geometric and representation-theoretic problems into computational frameworks, including custom algebraic solvers.

Machine Learning (PINNs): Applied Physics-Informed Neural Networks (PINNs) to approximate solutions for non-linear geometric equations (e.g., the Nirenberg problem). Contributed to the *AInstein* project group at the London Geometry and Machine Learning Summer School.

Scholarships

09/2024–12/2024 **Research Scholarship** by *Deutscher Akademischer Austauschdienst*, Bonn
Awarded to participate in the SLMath Semester Program, Berkeley

10/2020–09/2021 **Private Scholarship** by *Dr. med. Alexander Liesenfeld,* Amöneburg
Recommended by the Department of Mathematics and Computer Science, Philipps-University of Marburg

Professional Experience

Academic Positions

From 10/2022 **Research Associate**, *Philipps-University of Marburg*, Germany.

Research Stays

08/2024–12/2024 **Program Associate**, *Simons Laufer Mathematical Sciences Institute*, Berkeley, USA.

Additional Professional Experience

10/2018–07/2024 **Technical Assistant**, *Marburg Ion Beam Therapy Center*, Germany.

Presentations

Lecture Series and Invited Talks

07/2026 **Spectral Geometry of Deformed Normal Homogeneous Spaces**, *Virtual Seminar on Geometry with Symmetries*, Online.

01/2026 **AI-Assisted Research in (pure) Math**, *Graduate Training Workshop, INI*, Cambridge, UK.

09/2025 **Researching with Thinking Models: AI-Assisted Paths in Mathematics**, *The Annual 2025 ÖMG-DMV Meeting*, Linz, Austria.

02/2024 **Spectral Geometry (Five-Day Lecture Series)**, *University of Malaga*, Spain (invited).

11/2023 **Spectral Geometry of Aloff-Wallach Spaces**, *Oberseminar Geometrie und Topologie, University of Stuttgart*, Germany.

Conference and Seminar Presentations

AI in Spectral Geometry

04/2026 *An AI-Assisted Proof in Spectral Geometry*, GSTW07 Workshop, INI, Cambridge, UK.

Spectral Geometry on Naturally Reductive Spaces

09/2025 *The Annual 2025 ÖMG-DMV Meeting*, Linz, Austria.

09/2025 *Differential Geometry & Applications, Masaryk University*, Brno, Czech Republic.

10/2024 *Graduate Seminar, Simons Laufer Mathematical Sciences Institute*, Berkeley, USA.

06/2024 *41. Süddeutsches Kolloquium über Differentialgeometrie, Goethe University Frankfurt.*

03/2025 **The Dirac Operator with Skew Torsion on Naturally Reductive Spaces**, *Lie Theory in Geometry, Algebra and Analysis, National University of Cordoba*, Argentina.

Geometry of Aloff-Wallach Spaces

08/2023 *Prospects in Geometry and Global Analysis, Castle Rauischholzhausen*, Germany.

05/2023 *Spinorial & Octonionic Aspects of G_2 and Spin(7) Geometry, BIRS*, Banff, Canada.

Poster Presentations

Geometry of Aloff-Wallach Spaces

08/2023 *Workshop on Curvature and Global Shape, University of Münster*, Germany.

08/2023 *Prospects in Geometry and Global Analysis, Castle Rauischholzhausen*, Germany.

University Teaching at Philipps-University of Marburg

Co-Lecturer AI in Mathematics Education (Summer 2026). Co-designing and co-leading a seminar focusing on didactic methods for integrating artificial intelligence into mathematics teaching.

Lead Tutor Foundations of Math (Winter 2025/26), Analysis 1 (Summer 2025, 2023) Complex & Vector Analysis (Summer 2024, 2022), PDEs (Summer 2024), Lie Groups & Lie Algebras (Winter 2023/24), Differential Geometry 1 (Winter 2022/23).

Interactive Workshops Co-organized and led 90-minute interactive sessions in Analysis 1 (Summer 2025, 2023). Provided hands-on guidance to students, teaching them how to construct and structure mathematical proofs.

Tutor Functional Analysis (Winter 2021/22), Complex Analysis & Vector Analysis (Summer 2021).

Selected Conference, Workshop, and School Participation

08/2025 **AI Meets Mathematics Workshop**, *Imperial College*, London.

07/2025 **London Geometry and Machine Learning Summer School**, *Imperial College*.

02/2024 **Mathematical Supergravity**, *UNED - Madrid*.

10/2023 **Workshop on Einstein Manifolds**, *University of Stuttgart*.

03/2023 **Seminar Sophus Lie**, *Friedrich-Alexander-Universität*, Erlangen.

11/2022 **Oberwolfach Graduate Seminar: Geometry and Topology of Compact Homogeneous Spaces**, *Castle Będlewo*, Poland.

08/2022-09/2022 **Spaces, Structures, Symmetries**, *Università degli Studi di Bari Aldo Moro*, Bari, Italy.

06/2022 **Seminar Sophus Lie**, *Sophus Lie Conference Center*, Nordfjordeid, Norway.

10/2021 **Special Geometries on Riemannian Manifolds**, *UQAM*, Montreal.

Voluntary activities

10/2019–02/2022 **Moderator and Member of the Participation Advisory Board**, *Department 72 - Citizen Participation, Magistrate*, Marburg.
Advised on projects to encourage citizen participation in regional democratic decisions

Languages

German (native), English (fluent), Latin (basic knowledge)

References

Prof. I. Agricola Full Professor of Differential Geometry, University of Marburg, Germany.

Prof. E. Lauret Indep. Researcher (CONICET)/Assoc. Prof, Universidad Nacional del Sur, Argentina.